

indigenous ingenuity

Technology has allowed us to go
marching on, on, on!

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they say necessity is the mother of invention. Maybe invention is too strong a word here, but due to various historical, geopolitical, and demographic reasons – not to mention problems unique to India itself – our country has often had to take the road less travelled towards smart solutions designed to overcome critical challenges. And technology has always been at the forefront of these initiatives.

While you'll read about other aspects of India's technology journey and stalwarts who were instrumental in making their mark and leaving a lasting legacy of their work through innovations in technology elsewhere in this cover story, this particular article is dedicated to highlight some of our nation's ingenious achievements. Either alone, or in close partnership with a handful of international partners, how India tried to find a way to get things done. To go beyond call of duty, at times, even when completely isolated in an increasingly globalized world. Against all odds, contrary to popular opinion, through technology we not only got by, but actually flourished. This is that story.

THORIUM-BASED NUCLEAR REACTORS

In a world that's rapidly hurtling towards a mass energy crisis, renewable energy remains a pipedream that's only in its infancy. According to a report from the Central Electricity Authority, only 2 per

cent of India's electricity is generated from nuclear power (the cleanest source of non-renewable energy), whereas coal dependency constitutes a whopping 61 per cent thereby adversely affecting our climate and ecology. We have rightly identified the need to increase our nuclear power output, and leading the charge on indigenously developing thorium-based reactors.

Why is this a big deal? Simply put, getting nuclear power isn't as easy as generating electricity by burning coal. Nuclear



energy is derived from sensitive radioactive elements such as uranium, plutonium, thorium, etc., and importing technology and radioactive material either needs inter-governmental bilateral trade treaties or access to the NSG (Nuclear Suppliers Group) – a list of nuclear supplier countries that seek to prevent nuclear weapons proliferation by controlling and restricting the export of nuclear material and technologies. Since India isn't inside the NSG, we aren't allowed to buy nuclear material from countries that are part of the NSG. And for

whatever naturally occurring uranium and plutonium we have, we've been importing reactors built by other nations (Canada, Russia, US, etc.), which come with international safeguards.

To find an ingenious solution to this problem, India's embarked on a path of nuclear self-reliance. We don't have huge deposits of uranium, but we have the largest reserve of thorium in the world, and we're developing thorium-based reactors which are completely unique and home-grown.

The long-term objective of attaining self-sustainable nuclear energy starts with developing an advanced heavy-water thorium cycle. There are three stages of achieving this, by treating the thorium in different reactors. We've successfully completed two stages of this objective, and soon to implement stage three of the program where Advanced Heavy Water Reactors (AHWRs) will drive thorium-plutonium fuel to the U-233 stage which will enable a self-sustainable fissile material. We are also creating Molten Salt Breeder Reactors (MSBR) meant for large-scale deployment of thorium-based nuclear power plants all around the country.

Currently, nuclear power only contributes 5780 MW of 303,118 MW, and it's expected to grow to 6,700 MW (by nearly 1,000 MW) by the end of this year itself. By not having to import both the nuclear material and the power plants needed to generate electricity, thorium-based nuclear energy generation is going to help the